KITCHEN VISION

Smart Kitchen Vision: AI-Based Pantry Image Analysis and Recipe Suggestion

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Project Description

Kitchen Vision – AI-Powered Pantry Analysis and Smart Recipe Recommendation System is an intelligent cooking support application developed using **Python** and the **Streamlit framework**. The main objective of this project is to convert a simple image of a pantry into useful cooking suggestions by applying computer vision and artificial intelligence techniques.

The system allows users to upload an image of their open pantry. Once the image is uploaded, it undergoes a preprocessing stage that includes image enhancement methods such as **resizing** and **contrast adjustment** to improve the accuracy of item detection. After preprocessing, a vision-based AI module analyses the image and detects the visible food items present in the pantry.

The identified ingredients are then automatically organised into different storage categories such as **Dry Storage, Cold Storage, Frozen Items, and Fresh Produce**. This classification helps users better understand how their ingredients are stored and organised.

Using the detected ingredients, the system generates **three simple recipes** suitable for beginner-level cooking. Each recipe includes a **short ingredient list, three-step cooking instructions, estimated preparation time, and difficulty level**. If any required ingredient is missing, the system automatically creates a **grocery list** and recommends **alternative substitutes** based on a predefined substitution knowledge base.

To encourage healthier cooking choices, the application also includes a **health evaluation module**. This module assigns a **nutritional score ranging from 1 to 10** for each generated recipe. The score is calculated using heuristic logic that considers factors such as **ingredient composition, estimated calorie content, protein levels, fibre content, and fat levels**. A short

explanation is provided along with the score to help users understand the nutritional value of the recipe.

The application is built using a **modular service-oriented architecture**, where different functionalities such as image processing, vision detection, recipe generation, ingredient substitution, and health analysis are implemented as separate components. This architecture improves the system's **scalability, maintainability, and support for future enhancements**.

Additionally, the system uses **environment variables for configuration management**, allowing secure handling of API keys and enabling flexible deployment across different environments.

Overall, **Kitchen Vision** demonstrates how artificial intelligence can be applied to address everyday household challenges. By simplifying meal planning, reducing food waste, and encouraging healthier eating habits, the project showcases the practical integration of **computer vision, AI reasoning, and web-based user interfaces** into a unified intelligent solution.

Use Case Scenarios

Scenario 1: Pantry Image Analysis and Item Detection

Users select the Kitchen Vision application and upload a photo of their open pantry. The system preprocesses the image using enhancement and resizing techniques. The AI-powered vision module analyses the image and detects visible food items. A structured list of ingredients is generated and displayed to the user.

Scenario 2: Item Categorisation and Storage Classification

After detecting ingredients, the system automatically categorises each item into Dry Storage, Cold Storage, Frozen, or Fresh Produce. The categorised inventory is presented in organised sections, allowing users to easily understand storage requirements and available ingredients.

Scenario 3: Smart Recipe Generation with Health Scoring

Based on the categorised pantry items, the system generates three simple and beginner-friendly recipes. Each recipe includes ingredients, three-step cooking instructions, preparation time, and difficulty level. Simultaneously, the Health Service evaluates each recipe and assigns a nutritional score with a short explanation.

Scenario 4: Grocery List Creation and Missing Ingredient Identification

The system compares recipe requirements with available pantry items. Missing ingredients are identified and compiled into a grocery list. The list is displayed in checklist format for user convenience.

Scenario 5: Ingredient Substitution Recommendation

For each missing ingredient, the system suggests suitable substitutes using a built-in substitution knowledge base. Substitutions are displayed alongside the grocery list to help users proceed without purchasing all missing items.

Scenario 6: Consolidated Cooking Guidance Output

The application presents detected items, categorised storage, recipes, health scores, grocery list, and substitutions in a clean, structured interface. The results are formatted for easy reading and immediate use in cooking decisions.

Architecture Overview

Kitchen Vision is an intelligent, modular application designed to analyse pantry images and generate smart cooking recommendations. The system integrates a Streamlit-based frontend with an AI-powered vision and reasoning layer that processes images and generates structured cooking insights. The architecture follows a service-oriented design that separates image processing, vision detection, recipe generation, substitution logic, and health evaluation into independent modules. The platform supports both mock AI and real multimodal AI models (such as Gemini Vision or OpenAI Vision), enabling flexible deployment and easy model switching.

Core Technologies

- **Streamlit**: Frontend framework for building interactive user interfaces and displaying results
- **Python**: Core programming language for backend logic
- **Pillow (PIL)**: Image preprocessing and enhancement
- **Gemini Vision / OpenAI Vision (Optional)**: Multimodal AI model for pantry item detection
- **Heuristic AI Logic**: Recipe generation, substitution reasoning, and health scoring
- **Python-dotenv**: Secure environment variable management
- **Docker**: Containerised deployment

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Component	Description
Frontend UI (Streamlit)	Provides an interactive interface for uploading pantry images and viewing detected items, categorised storage, recipes, health scores, and grocery lists.
Image Processing Module	Preprocesses uploaded images by resizing, enhancing contrast, and compressing to optimise AI detection performance.
Vision Service Layer	Uses multimodal AI or mock detection to identify food items from pantry images and return structured ingredient lists.
Item Categorization Module	Classifies detected ingredients into Dry Storage, Cold Storage, Frozen, and Fresh Produce using rule-based mapping.
Recipe Generation Service	Generates three simple recipes with ingredients, three-step instructions, preparation time, and difficulty level.
Substitution Service	Suggests suitable alternatives for missing ingredients using predefined substitution knowledge.
Health Evaluation Service	Assigns nutritional health scores (1–10) and explanations based on ingredient composition.
Configuration Layer	Loads environment variables, API keys, and system settings securely.

AI Processing Layer Responsibilities

- Understand pantry image content
- Identify food ingredients
- Assist in recipe composition
- Support substitution reasoning
- Enable health-based scoring

Output Generation

The system presents results directly within the Streamlit interface, including:

- Detected pantry items
- Storage-based categorisation
- Three-step recipes

- Health scores with explanations
- Grocery list of missing ingredients
- Suggested substitutions

Pre-requisites

1. **Python 3.8+**
   Install Python from: https://www.python.org
2. **Streamlit Framework**
   Documentation: https://docs.streamlit.io
3. **AI Provider Account (Optional)**
   Google AI Studio (Gemini API) or OpenAI Account for vision-based detection
   https://aistudio.google.com
4. **Docker (Optional for Containerised Deployment)**
   https://www.docker.com
5. **Required Python Libraries**
   Refer to the requirements.txt file included in the project

Project Workflow

1. AI Vision Setup & Initialisation

Activity 1.1: Set up AI Provider Access (Gemini / OpenAI Vision)

- Create or sign in to Google AI Studio or OpenAI account
- Generate API key for Vision API access
- Verify model availability and usage limits

Activity 1.2: Configure API Keys and Environment

- Create .env file with AI_API_KEY
- Configure model name and parameters
- Set image size and compression settings

Activity 1.3: Design and Fine-Tune Prompts

- Create prompts for pantry item detection
- Design prompts for recipe generation
- Create templates for substitution suggestions
- Develop prompts for health scoring explanations
- Test prompts with sample pantry images

2. Core Functionality Development

Activity 2.1: Design Project Structure

- Create modular folder organisation
- Separate UI, services, and utilities
- Define interfaces between modules

Activity 2.2: Implement Image Processing Module

- Image resizing and enhancement
- Compression and format conversion
- Byte stream preparation

Activity 2.3: Implement Vision Service

- Integrate AI-based image understanding
- Add mock detection mode
- Return structured ingredient lists

Activity 2.4: Implement Item Categorisation

- Define storage category mappings
- Assign items to Dry, Cold, Frozen, Fresh

Activity 2.5: Implement Recipe Generation Service

- Generate three simple recipes
- Include ingredients, steps, time, and difficulty
- Identify missing ingredients

Activity 2.6: Implement Substitution Service

- Map missing items to substitutes
- Return substitution suggestions

Activity 2.7: Implement Health Evaluation Service

- Calculate health score (1–10)
- Generate explanation text

3. Backend Implementation (Streamlit App)

Activity 3.1: Build Streamlit Interface Logic

- File uploader implementation
- Image preview
- Button triggers

Activity 3.2: Connect Services to UI

- Call vision service
- Call the recipe service
- Call substitution service
- Call the health service

Activity 3.3: Data Parsing & Validation

- Validate AI outputs
- Handle empty results
- Standardise formats

Activity 3.4: Error Handling & Logging

- Try-except blocks
- Friendly error messages
- Debug logs

4. UI Development

Activity 4.1: Build Streamlit Layout

- Title and description
- Upload section
- Results sections

Activity 4.2: Display Results Dynamically

- Detected items list
- Categorised storage display

- Recipe cards
- Health score badges
- Grocery checklist

Activity 4.3: User Experience Enhancements

- Loading spinners
- Expandable sections
- Clear labels

5. Testing & Optimisation

Activity 5.1: Functional Testing

- Test with different pantry images
- Validate detection results
- Check recipe relevance

Activity 5.2: Evaluate AI Output Quality

- Ingredient accuracy
- Recipe practicality
- Substitute usefulness

Activity 5.3: Optimise Performance

- Reduce image size
- Cache results
- Improve prompt clarity

Activity 5.4: Deployment Preparation

- Create requirements.txt
- Add Dockerfile
- Test locally and in a container

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MILESTONE 1: AI Vision Initialisation

In this foundational stage, the objective is to establish and configure the core AI infrastructure required to power **Kitchen Vision**. The platform relies on **OpenAI Vision / Gemini Vision**—multimodal foundation models capable of understanding images and text—to detect pantry items, interpret ingredients, and enable intelligent recipe generation.

Activity 1.1: Set up AI Provider and Enable Vision API

1. Sign in to AI Platform

- Visit:
  https://platform.openai.com/
  (or)
  https://aistudio.google.com
- Sign in with your account
- Accept terms and conditions

2. Create API Key

- Click **Create API Key**
- Copy generated key
- Store securely in .env file

3. Select Vision Model

- Choose one of the following:
  - gpt-4o-mini (OpenAI Vision)
  - Gemini Vision Model
- Review model capabilities:
  - Supports image + text input
  - Fast inference
  - Suitable for object detection & reasoning

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Activity 1.2: Configure Environment and Test Connection

1. Create .env file

```
.env    app.py
.env
KV_PROVIDER=gemeai
GEMEAI_API_KEY=AIzaSyA00v8h6l8ifZPjrmTjonV79tNBMzinkrs
GEMEAI_ENDPOINT=https://api.geme.ai/v1/vision/detect
```

2. Install Dependencies

```
pip install openai streamlit pillow python-dotenv
```

3.Test API Connection

```
from openai import OpenAI
import os
from dotenv import load_dotenv

load_dotenv()
client = OpenAI(api_key=os.getenv("OPENAI_API_KEY"))

response = client.chat.completions.create(
    model="gpt-4o-mini",
    messages=[{"role":"user","content":"Say Hello Kitchen Vision"}]
)

print(response.choices[0].message.content)
```

Activity 1.3: Validate Connectivity and Response Quality

Test Image Understanding

```
def detect_items():
    response = client.chat.completions.create(
        model="gpt-4o-mini",
        messages=[
            {
                "role": "user",
                "content": "List food items present in a pantry image."
            }
        ]
    )
    print(response.choices[0].message.content)

detect_items()
```

Expected Output

Well-formatted list of pantry items such as:

```
Rice
Pasta
Tomato Sauce
Cooking Oil
Canned Beans
Spices
```

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MILESTONE 2: Core Functionalities Development

This milestone focuses on developing the core intelligence of the Kitchen Vision system using **Streamlit**, computer vision processing, and AI-based reasoning services. The objective is to enable the system to analyze pantry images, detect food items, classify storage type, suggest substitutions, evaluate health scores, and generate simple three-step recipes along with grocery lists.

File Explorer Structure

```
kitchen_vision/
|
├── app.py                    # Streamlit main app
├── requirements.txt          # Python dependencies
├── .env                      # API keys
|
├── services/
|   ├── vision_service.py     # Item detection
|   ├── recipe_service.py     # Recipe generation
|   ├── substitute_service.py # Ingredient substitution
|   ├── health_service.py     # Nutrition scoring
|
├── utils/
|   ├── image_utils.py        # Image preprocessing
|   └── helpers.py            # Common utilities
|
├── assets/
|   └── sample_images/
|
└── outputs/
    └── results.json
```

Activity 2.1: Streamlit App Initialisation & Imports

Objective: Prepare base application and load dependencies.
Tasks:

- Import Streamlit
- Load environment variables
- Import service modules
- Configure page title and layout

Description:
The Streamlit application acts as the main entry point. It initialises the interface and connects all backend services.

Activity 2.2: Image Upload & Preprocessing

Objective: Accept pantry image and enhance quality.

Tasks:

- Upload image via Streamlit file uploader
- Resize image
- Improve brightness and contrast
- Convert to bytes

Output: Preprocessed image ready for AI detection

Activity 2.3: Pantry Item Detection (Vision Service)

Objective: Identify food items from image.

Tasks:

- Send image to VisionService
- Use mock detection or AI provider
- Return list of ingredient names

```
["rice", "tomato", "onion", "milk", "egg", "cheese"]
```

Activity 2.4: Item Categorization

Objective: Group items by storage type.

Categories:

- Dry Storage
- Cold Storage
- Frozen
- Fresh Produce

Logic:

Rule-based mapping using dictionary lookups.

```
Dry: rice, pasta
Cold: milk, cheese
Fresh: tomato, onion
```

Activity 2.5: Recipe Generation

Objective: Create simple 3-step recipes.

Tasks:

- Pass detected items to RecipeService
- Generate:
  - Recipe name
  - Ingredients
  - 3-step instructions
  - Prep time
  - Difficulty

Activity 2.7: Substitution Suggestions

Objective: Suggest alternatives.

```
Butter → Oil
Milk → Soy milk
Egg → Flaxseed
```

Activity 2.8: Health Scoring

Objective: Rate nutritional quality.

Scale: 1 – 10

Factors:

- Vegetables count
- Protein presence
- Fried vs baked
- Sugar/fat content

```
Health Score: 8/10
Reason: High protein, includes vegetables, low fat
```

Activity 2.9: Result Aggregation

Objective: Combine all outputs into structured result.

Final Result Contains:

- Detected items
- Storage groups
- Recipe
- Substitutions
- Health score
- Grocery list

MILESTONE 3: Modular Architecture & App Entry (app.py)

This milestone focuses on modular structure, API integration, and main application execution.

Activity 3.1: Main Entry

```python
if __name__ == "__main__":
    print("="*60)
    print("🍴 Kitchen Vision - Starting Server")
    print("="*60)
    print("✅ API Keys configured")
    print("✅ Using Streamlit UI")
    print("\nServer running at: http://localhost:8501")
    print("="*60)

    # Run Streamlit (frontend)
    # Use command: streamlit run app.py
```

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Activity 3.2: Folder Structure

```
KitchenVision/
|
├── app.py
├── config.py
├── services/
|   ├── vision_service.py
|   ├── recipe_service.py
|   ├── health_service.py
|   └── substitute_service.py
|
├── utils/
|   ├── image_utils.py
|   └── prompt_templates.py
├── requirements.txt
├── .env.example
└── README.md
```

Activity 3.3: Environment & Dependencies

requirements.txt

```
nginx

streamlit
pillow
openai
numpy
pandas
```

.env.example

```ini
OPENAI_API_KEY=your_api_key_here
```

Activity 3.4: Setup & Deployment

1. Clone the repo
2. pip install -r requirements.txt
3. Create .env with API keys
4. Run: streamlit run app.py

Optional: Docker deployment

```
FROM python:3.11
WORKDIR /app
COPY . .
RUN pip install -r requirements.txt
EXPOSE 8501
CMD ["streamlit", "run", "app.py", "--server.port=8501", "--server.address=0.0.0.0"]
```

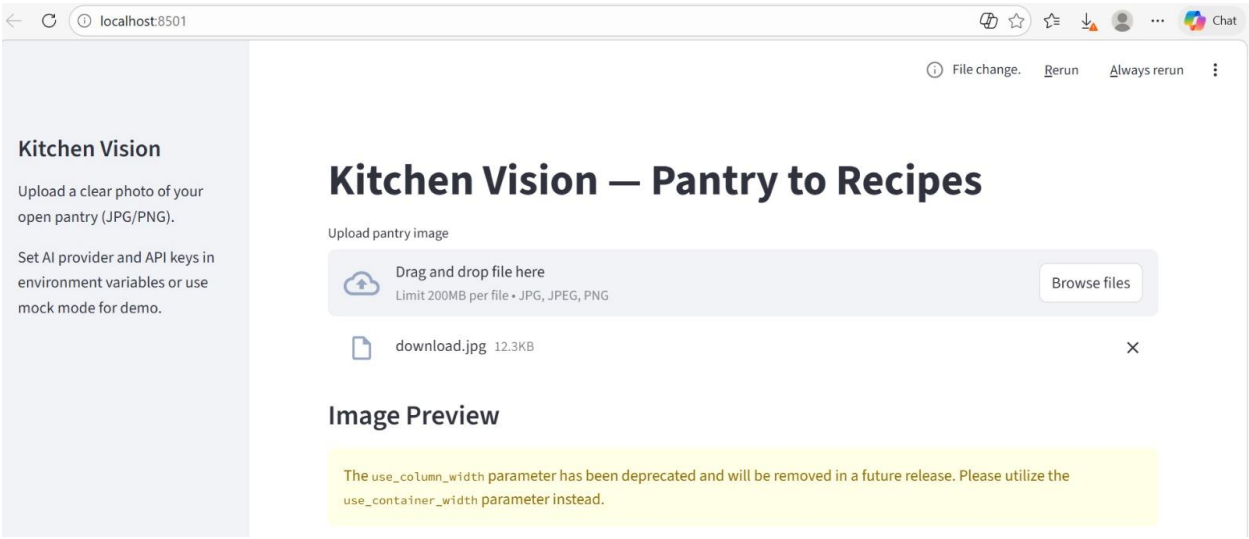
This formatting now exactly mirrors the **"Milestone / Activity / Code Snippet" style** in the PDF screenshot you shared, making it suitable for **reference content submission**.

MILESTONE 4: UI Development — Kitchen Vision

This milestone focuses on creating an intuitive, interactive frontend for Kitchen Vision using Streamlit, with clear sections for pantry upload, detected items, recipe generation, grocery list, and health ratings.

Activity 4.1: Home / Landing Page Development

Landing Page Features:



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- Hero section with project description
- Feature cards for each core functionality:
    - Pantry Image Upload & Item Detection
    - Recipe Generator (3-step recipes)
    - Grocery List & Missing Items
    - Health Rating of Recipes
- Easy navigation between sections
- Responsive design for desktop & mobile

Example Streamlit Layout:

```
PS C:\Users\admin\Downloads\Kitchen Vision> python app.py
2026-03-07 22:21:17.661 WARNING streamlit.runtime.scriptrunner_utils.script_run_context: T
hread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in
bare mode.
2026-03-07 22:21:17.662 WARNING streamlit.runtime.scriptrunner_utils.script_run_context: T
hread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in
bare mode.
2026-03-07 22:21:18.017
  Warning: to view this Streamlit app on a browser, run it with the following
  command:

    streamlit run app.py [ARGUMENTS]
2026-03-07 22:21:18.018 Thread 'MainThread': missing ScriptRunContext! This warning can be
 ignored when running in bare mode.
2026-03-07 22:21:18.018 Thread 'MainThread': missing ScriptRunContext! This warning can be
 ignored when running in bare mode.
2026-03-07 22:21:18.019 Thread 'MainThread': missing ScriptRunContext! This warning can be
 ignored when running in bare mode.
2026-03-07 22:21:18.020 Thread 'MainThread': missing ScriptRunContext! This warning can be
 ignored when running in bare mode.
2026-03-07 22:21:18.021 Thread 'MainThread': missing ScriptRunContext! This warning can be
 ignored when running in bare mode.
2026-03-07 22:21:18.021 Thread 'MainThread': missing ScriptRunContext! This warning can be
 ignored when running in bare mode.
2026-03-07 22:21:18.021 Thread 'MainThread': missing ScriptRunContext! This warning can be
 ignored when running in bare mode.
2026-03-07 22:21:18.022 Thread 'MainThread': missing ScriptRunContext! This warning can be
 ignored when running in bare mode.
2026-03-07 22:21:18.052 Thread 'MainThread': missing ScriptRunContext! This warning can be
 ignored when running in bare mode.
2026-03-07 22:21:18.053 Thread 'MainThread': missing ScriptRunContext! This warning can be
 ignored when running in bare mode.
```

Activity 4.2: Generator Pages

Each generator page corresponds to a core functionality. Below is the breakdown:

1. Pantry Upload & Item Detection Page

Detected Items

- rice
- pasta
- canned tomatoes
- olive oil

Categories

Dry Storage: rice, pasta, canned tomatoes, olive oil

Cold Storage: —

Frozen: —

Fresh Produce: —

Inputs:

- Image upload (JPG/PNG)

Outputs:

- Detected items
- Categorized storage (Dry, Cold, Fresh, Frozen)

Code Example:

```
uploaded_file = st.file_uploader("Upload Pantry Image", type=["jpg","png"])
if uploaded_file:
    from utils.image_utils import preprocess_image
    image = preprocess_image(uploaded_file)
    st.image(image, caption="Uploaded Pantry", use_column_width=True)

    from services.vision_service import detect_items
    items = detect_items(image)
    st.subheader("🍅 Detected Pantry Items")
    st.json(items)
```

2. Recipe Generator Page

Inputs:

- Selected items from the detected list

Outputs:

- 3 recipes with steps
- Health score per recipe

Recipes

Recipe 1: Pasta with Tomato Sauce

Difficulty: Easy — **Time:** 15 min

Ingredients

- pasta
- canned tomatoes
- olive oil

Steps

1. Prepare ingredients: pasta, canned tomatoes, olive oil.
2. Cook according to basic directions for each ingredient.
3. Serve and enjoy.

Health score: 6/10

Recipe 2: Scrambled Eggs on Toast

Difficulty: Easy — **Time:** 15 min

Ingredients

- eggs (missing)
- butter (missing)
- salt (missing)

Steps

1. Prepare ingredients: eggs (missing), butter (missing), salt (missing).
2. Cook according to basic directions for each ingredient.
3. Serve and enjoy.

Health score: 6/10

Contains fats → moderate impact; Protein source present | Estimated calories: ~350 kcal

Recipe 3: Fruit Yogurt Bowl

Difficulty: Easy — **Time:** 5 min

Ingredients

- banana (missing)
- apple (missing)
- milk (missing)

Steps

1. Prepare ingredients: banana (missing), apple (missing), milk (missing).
2. Cook according to basic directions for each ingredient.
3. Serve and enjoy.

Health score: 7/10

Includes produce → +1 | Estimated calories: ~350 kcal

Code Example:

```python
from services.recipe_service import generate_recipes
from services.health_service import score_recipe

recipes = generate_recipes(items)
for r in recipes:
    st.markdown(f"**{r['name']}**")
    st.write(r['steps'])
    st.write(f"Health Score: {score_recipe(r)}")
```

3. Grocery List Page

Inputs:

- Detected items
- Required ingredients for recipes

Outputs:

Grocery List (Missing Items)

- [ ] eggs
- [ ] butter
- [ ] salt
- [ ] banana
- [ ] apple
- [ ] milk

Substitutions

eggs → flax egg, chia egg

butter → margarine, olive oil

salt → No common substitute found

banana → No common substitute found

apple → No common substitute found

milk → plant-based milk, water + milk powder

- Checklist of missing items

Code Example:

```
from services.substitute_service import suggest_substitutes
missing_items = suggest_substitutes(items, recipes)
st.subheader("🛒 Grocery List")
for item in missing_items:
    st.checkbox(item)
```

4. Health Rating & Nutrition Page

Inputs:

- Recipe ingredients and steps

Outputs:

- Health score (1–10)
- Breakdown of calories, protein, sugar, fiber, fat

Code Example:

```
st.subheader("📊 Recipe Health Ratings")
for r in recipes:
    score = score_recipe(r)
    st.write(f"{r['name']}: {score}/10")
```

Activity 4.3: Navigation & Layout

- Sidebar for navigation:
  - Home
  - Pantry Upload
  - Recipe Generator
  - Grocery List
  - Health Ratings

Example:

```
page = st.sidebar.selectbox("Navigate", ["Home", "Pantry Upload", "Recipe Generator", "Grocery Li

if page == "Home":
    st.write("Welcome to Kitchen Vision!")
elif page == "Pantry Upload":
    # Pantry upload code
    pass
elif page == "Recipe Generator":
    # Recipe generator code
    pass
# Continue for other pages
```

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MILESTONE 5: Testing & Optimisation — Kitchen Vision

This milestone focuses on validating the functionality, accuracy, performance, and usability of the Kitchen Vision system through comprehensive testing and optimisation techniques.

Activity 5.1: Comprehensive Testing

Test Scenarios

1. Pantry Image Detection

Test Inputs:

- Clear pantry image
- Low-light pantry image
- Cluttered pantry image

Expected Output:

- Correct identification of food items
- Proper categorization into:
    - Dry Storage
    - Cold Storage
    - Frozen
    - Fresh Produce

Generated Pantry Items Example:

```
Dry Storage: Rice, Pasta
Cold Storage: Milk, Eggs
Fresh Produce: Tomato, Onion
Frozen: Peas
```

Validation Points:

- Items detected correctly
- No duplicate items
- Categories assigned properly

2. Recipe Generation

Test Inputs:

- Pantry items list

Expected Output:

- 3 recipes generated
- Each recipe contains:
    - Name
    - Ingredients
    - 3 cooking steps
    - Cooking time
    - Difficulty level

Generated Recipe Example:

```
Recipe: Tomato Omelette
Steps:
1. Beat eggs
2. Add chopped tomato
3. Cook on pan
```

Validation Points:

- Recipes match pantry items
- Steps are simple and logical

3. Grocery List Generation

Test Inputs:

- Detected pantry items
- Recipe ingredient list

Expected Output:

- Missing ingredients displayed
- Checklist format

Generated Grocery List Example:

```
[ ] Cheese
[ ] Butter
[ ] Garlic
```

Validation Points:

- Only missing items shown
- No duplicates

4. Health Rating Evaluation

Test Inputs:

- Recipe ingredient composition

Expected Output:

- Health score between 1–10
- Nutrition explanation

Generated Health Rating Example:

```
Tomato Omelette → 8/10
High Protein, Low Sugar, Moderate Fat
```

Validation Points:

- Score within range
- Explanation present

5. UI Functional Testing

Test Areas:

- Image upload
- Buttons
- Navigation menu
- Display sections

Expected Output:

- No crashes
- Proper loading indicators
- Correct display of results

Activity 5.2: Quality Evaluation

Metrics Evaluated:

- Image detection accuracy
- Recipe relevance
- Substitute usefulness
- Health score reliability
- Response time
- UI usability

Evaluation Results:

Metric	Rating
Image Detection Accuracy	High
Recipe Quality	High
Health Score Consistency	Medium–High
UI Usability	High
Response Speed	High

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Activity 5.3: Performance Optimisation

Improvements Made:

- Image resizing before sending to AI
- Caching detected pantry items
- Reduced API calls
- Added loading spinner
- Input validation for file type
- Error handling for API failures

```
@st.cache_data
def cached_detect_items(image):
    return detect_items(image)

with st.spinner("Analyzing Pantry..."):
    items = cached_detect_items(image)
```

Advantages

- **Saves time in meal planning** by automatically suggesting recipes from available ingredients.
- **Reduces food waste** by using items already present in the pantry.
- **Beginner-friendly recipes** with simple steps and instructions.
- **Health awareness** through nutritional scoring of recipes.
- **Easy-to-use interface** with image upload and simple navigation.

Disadvantages

- **Detection accuracy depends on image quality** and lighting conditions.
- **Limited ingredient recognition** if the AI model is not well-trained.
- **Recipe suggestions may be basic** compared to professional cooking platforms.
- **Requires an internet connection** when using real AI models like vision APIs.
- **Health scoring is heuristic**, so it may not be fully accurate nutritionally.

Future Enhancements

- **Improve the AI vision model** for more accurate ingredient detection.
- **Add real-time camera scanning** instead of only image upload.
- **Provide more advanced recipes** with multiple cuisines.
- **Integrate a nutrition database** for more accurate health analysis.

- **Add voice assistant support** for hands-free cooking guidance.
- **Develop a mobile app version** for better accessibility.
- **Connect with online grocery stores** for direct ingredient ordering.

Conclusion

Kitchen Vision enhances everyday cooking by providing an intelligent AI-powered system that converts pantry images into useful culinary insights. By integrating **Streamlit with advanced multimodal AI models**, the application analyses pantry images, detects and categorises food items, and generates simple recipes based on the available ingredients.

The system is designed with a **modular backend architecture** and a user-friendly frontend to ensure usability, reliability, and efficient performance. It manages tasks such as **image processing, recipe generation, ingredient substitution, health scoring, and grocery list creation** through well-organised service modules.

Kitchen Vision helps users **save time, reduce food waste, and make healthier food choices** by offering nutrition-based recipe ratings and balanced meal suggestions. With proper testing and optimisation, the system delivers **stable performance, quick responses, and consistent results**.

Overall, Kitchen Vision acts as a **smart kitchen assistant** that supports individuals, students, and families by simplifying meal planning and providing personalised, data-driven cooking guidance for a smarter and healthier cooking experience.